

Research Paper

The potential of pectin-based films enriched with bioactive components for strawberry preservation: A sustainable and innovative coating

Gulce Bedis Kaynarca^a, Deniz Damla Altan Kamer^b, Emel Yucel^b, Oylum Simal Yilmaz^b, Yasemin Henden^b, Elif Kaymaz^b, Tuncay Gumus^{b,*}

^a Department of Food Engineering, Faculty of Engineering, Kırklareli University, Kırklareli 39100, Turkey

^b Department of Food Engineering, Tekirdag Namik Kemal University, Tekirdag 59030, Turkey



ARTICLE INFO

Keywords:

Pectin
Grape pomace
Wine lees
Edible coatings
Strawberry preservation

ABSTRACT

Fruits that are characterized by their fragile architecture are less durable and are a major source of waste. Thus, research on increasing the durability of such foods remains relevant. Pectin is an inexpensive and edible coating material derived from natural resources that can be used to preserve strawberries. The aim of the current research was to fortify pectin with grape pomace extract (GPE) and wine lees extract (WLE) that are rich in bioactive components. Physicochemical, microbiological and functional properties of coated and uncoated strawberries were investigated during the storage period (0, 5, 10, 15, and 20th day). GPE and WLE-supported pectin coatings helped preserve the antioxidant capacity, phenolic, anthocyanin and ascorbic acid contents of strawberries. Coated samples WLE-enriched exhibited significantly less weight loss compared to other samples. WLE coating was effective in inhibiting yeast mould populations on strawberries. WLE-enriched coatings acted as a wall material to preserve bioactive components and delayed quality changes compared to GPE, allowing them to reach a storage period of 20 days at 4 °C.

1. Introduction

The global demand for fresh fruits and vegetables is growing; yet, these perishable products have a limited shelf life due to rapid biochemical, physiological, and microbiological deterioration that occurs during postharvest processing, storage, and transit (Kumar et al., 2023). Strawberry fruit has a low storage potential as a result of its high respiration and transpiration rates, susceptibility to fungal decay, and fragile structure (Matar et al., 2020).

Strawberry (*Fragaria ananassa*) fruit belongs to the Rosaceae family and is rich in flavonoids, fiber, vitamins, potassium and a wide range of phenolic acids such as hydroxycinnamic and hydroxybenzoic acids (Quarshi et al., 2023). Strawberries have a high vitamin C and folate content and a low calorie and glycemic index, which can prevent the development of Type-2 diabetes (Basu et al., 2016). In addition, their anti-inflammatory, anti-cancer and anti-microbial properties offer various health benefits (Hernández-Martínez et al., 2023). The global demand for strawberries has surged by over 80 % in recent decades resulting in an output of more than 9 million tons (Kaur and Kumar, 2022). Considering the high production and perishable nature of

strawberries, it is ecologically and economically important to extend the storage period to minimize waste. Freezing, gas regulation, irradiation, ultra-high pressure, and chemical methods can be used to reduce post-harvest respiration rate and water loss, maintain fruit firmness, and prevent microbial growth (Guo et al., 2024).

High operating requirements, cost, and chemically induced chronic food poisoning have increased research interest in innovative preservation methods (Guo et al., 2024). The current trend involves the utilization of natural coating materials and incorporation of natural additives such as plant extracts as environmentally friendly enhancers to improve the performance of coatings (De Bruno et al., 2023). Coatings made from low methoxy pectin (LMP) that is enhanced with natural antioxidants and/or antimicrobials were shown to be a successful substitute for synthetic wax-based coatings (Kumar et al., 2023).

Pectin, which is the methylated ester of polygalacturonic acid, finds extensive applications in the food, pharmaceutical, and cosmetic sectors. Many agricultural waste and food processing by-products are a low-cost and rich source of pectin. Pectin is classified as a prebiotic, serving as a rich source of dietary fiber that cannot be broken down by the human gastrointestinal system (Pholsin et al., 2024). Pectin is an

* Corresponding author.

E-mail address: tgumus@nku.edu.tr (T. Gumus).

<https://doi.org/10.1016/j.scienta.2024.113294>

Received 15 February 2024; Received in revised form 3 April 2024; Accepted 9 May 2024

Available online 15 May 2024

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advantageous natural resource that can be used in food packaging due to its abundance, coating ability, biodegradability, and bioavailability (Wu et al., 2023). Pectin can be classified into high methoxyl (HMP) and low methoxyl (LMP) according to the degree of methyl esterification. LMP is more widely used in coating technology as HMP has a high elastic modulus (Nastasi et al., 2022).

Flavonoids and phenolic acids with antioxidant, UV protection, and antimicrobial properties are of plant origin and can interact with plant cell wall (PCW) components, especially polysaccharides such as pectin (Siemińska-Kuczer et al., 2022). Upon reaching the intestine, the pectin-polyphenol complex formed by hydrogen and hydrophobic bonds can be fermented by bacteria. The fermented products are more beneficial for human health compared to free polyphenols (Méndez et al., 2023). Essential oils have been used in the structure of pectin-based coatings containing plant extracts as natural preservatives (Nastasi et al., 2022). The use of phenolic plant extracts containing high amounts of phenolic acids, flavonoids, anthocyanins, tannins, and polyphenols has been limited to green tea powder (Kang et al., 2007), cinnamon (Nimah, 2020), olive by-product (Bermúdez-Oria et al., 2019) and leaves (Kurek et al., 2021). The most common by-products of wine production are grape pomace (Antonić et al., 2020) and wine lees (Gümüş et al., 2024), which are rich sources of fiber, phenolic compounds, colorants, minerals and organic acids. These waste products raise environmental concerns due to their organic content and the potential for contamination of water resources if not managed properly. In addition, the disposal of these materials is expensive, highlighting a need for sustainable alternatives (Umsza-Guez et al., 2023).

The current global economy entails extensive imports and exports of food products between countries. Therefore, studies on extending the storage period of fresh products have become widespread. In addition, waste management and reduction of waste generation have become important issues within the scope of the green agreement. We have previously reported the extension of the preservation of strawberries using PVA-based films (Kaynarca et al., 2023). In the current study, we aimed to increase the preservation of strawberry with pectin, which is a low cost and sustainable source of natural coating material. The use of grape pomace and wine residue in pectin coating has not been reported previously. We examined the effects of grape pomace extract (GPE) and wine lees extract (WLE) on the quality parameters of strawberries during the storage period and thus presented an innovative and novel perspective to utilize these waste products as an environmentally friendly, sustainable, and nutritious coating material. In addition, we have examined how the taste, aroma, and color of the coating material resourced from the waste products can affect the commercial value of the berries.

2. Material and methods

2.1. Materials

The thermo-reversible low-methoxyl apple pectin (NH Nappage) was obtained from a local supplier (Global Pastacılık ve Pasta Malzemeleri Ltd.Şti.). Winery by-products, grape pomace (*Vitis vinifera* subsp. *vinifera* cv. Cabernet-Sauvignon) and wine lees were sourced from a boutique winery in Tekirdağ, Şarköy. Grape pomace, the residual material left after the extraction of juice from grapes, is a significant source of winery waste. Similarly, wine lees, the sediment accumulated during wine fermentation and aging, represent another underutilized winery by-product. Grape pomace and wine lees were dried in a drying oven at 35 °C, ground and extracted with 70 % ethanol (v/v) at a ratio of 1:5 (w/v). GPE and WLE were generated after evaporation of the ethanol and stored at −18 °C until use.

Strawberries of the same freshness, color, and size were gathered from a strawberry field (latitude 40°57'10.9" N, longitude 27°27'05.3" E) that was nearest to the city center of Tekirdağ, Turkey in July 2023. To minimize the degradation of strawberry quality, freshly collected

samples were coated on the same day and stored at 4.

2.2. Preparation of coating solutions and their application

5 % pectin solution (w/v) was slowly added to deionized water and stirred in a magnetic stirrer until complete hydration was achieved. Based on our previous investigations (Kaynarca et al., 2023), 1 % (v/v) aqueous extracts of WLE and GPE were added to the solution.

The strawberries were washed for 3 min in cold, sterile, distilled water, followed by 30 min of drying at 20 °C. The control sample (C) consisted of strawberries that had been submerged in distilled water for 3 min before being allowed to dry. The test samples were immersed in either Pectin (P) or Pectin+GPE (PGP) or Pectin+WLE (PWL) solutions for 2 min and drained on metal wires for 1 min. Immersion and draining process was repeated 3 times. The coated strawberries were left to dry for 1 hour. All samples were kept in polypropylene containers measuring 16×9×5 cm at 4 °C through the 20-day analytical period (Fig.1).

2.3. Bioactive compounds and antioxidant capacity of coating solutions

The total phenolic content of the solutions (P, PGP, PWL) was measured with a spectrophotometer (Shimadzu 02910, Tokyo, Japan) at 760 nm using the Folin-Ciocalteu reagent. The samples (40 µL) were diluted 6-fold with pure water, mixed with 200 µL of Folin-Ciocalteu reagent and incubated for 2 min. 600 µL of this solution was mixed with 75 g L^{−1} Na₂CO₃ and 760 µL distilled water by vortexing followed by measurement of the absorbance at 760 nm. The measurements were taken quickly after 1 hour of incubation in the dark (Singleton, 1999).

The antioxidant activity of the coating solutions was measured with 1,1-diphenyl-2-picrylhydrazyl (DPPH) and cupric-reducing antioxidant capacity (CUPRAC) assays. Methanolic extracts of the solutions were mixed with 600 µL of 1 mM DPPH (1,1-diphenyl-2-picrylhydrazyl) methanolic solution to a final volume of 6000 µL with followed by 30 min of incubation in the dark. The absorbance of the solution was measured at 517 nm and the IC₅₀ value for radical scavenging activity was calculated (mg L^{−1}). For the CUPRAC assay, 500 µL of the sample solution was mixed with 600 µL of distilled water, 1000 µL of 0.0075 M neocuproine (2,9-dimethyl-1,10-phenanthroline) in ethanol, and 1000 µL of 0.01 M copper II chloride. The mixture was incubated in the dark for an hour at 25 °C before measuring the absorbance at 450 nm. The data are reported in mM Trolox equivalent (mM TE) (Gümüş et al., 2024).

Total anthocyanin content (TAC) of the solutions was determined by the pH differential method (Wrolstad, 1993). The samples were extracted in methanol containing 2 % HCl at a ratio of 1:5 for 60 min. This process was repeated three times. The samples were then passed through a 0.45 µm filter and diluted in potassium chloride and sodium acetate buffer adjusted to pH 1.0 and 4.5. The absorbance was measured at 522 and 700 nm. The total anthocyanin content of the coating solutions was calculated in terms of malvidin-3-glucoside with a molecular weight of 493.5 and molar absorbance of (ε) 28000 L cm^{−1} mg^{−1}.

2.4. Effect of pectin-based coating on cold storage quality of strawberries

2.4.1. Weight loss

The weights of 10 strawberries randomly selected from each group of samples (C, P, PGP and PWL) were weighed on a digital balance (Isolab, Eschau, Germany) during the storage period. The weight loss (%) was calculated as the percentage of weight lost per original fruit weight.

2.4.2. pH and titratable acidity, and total soluble solid content

Titratable acidity (TA), pH and total soluble solid content (TSS) were determined according to Assn. of Official Analytical Chemists (AOAC 1990). TSS (%) was determined at 20 °C using a digital refractometer (ATAGO, Tokyo, Japan). TA was represented as the percentage (%) of citric acid. The acidity was measured with a pH meter (HANNA HI2002,

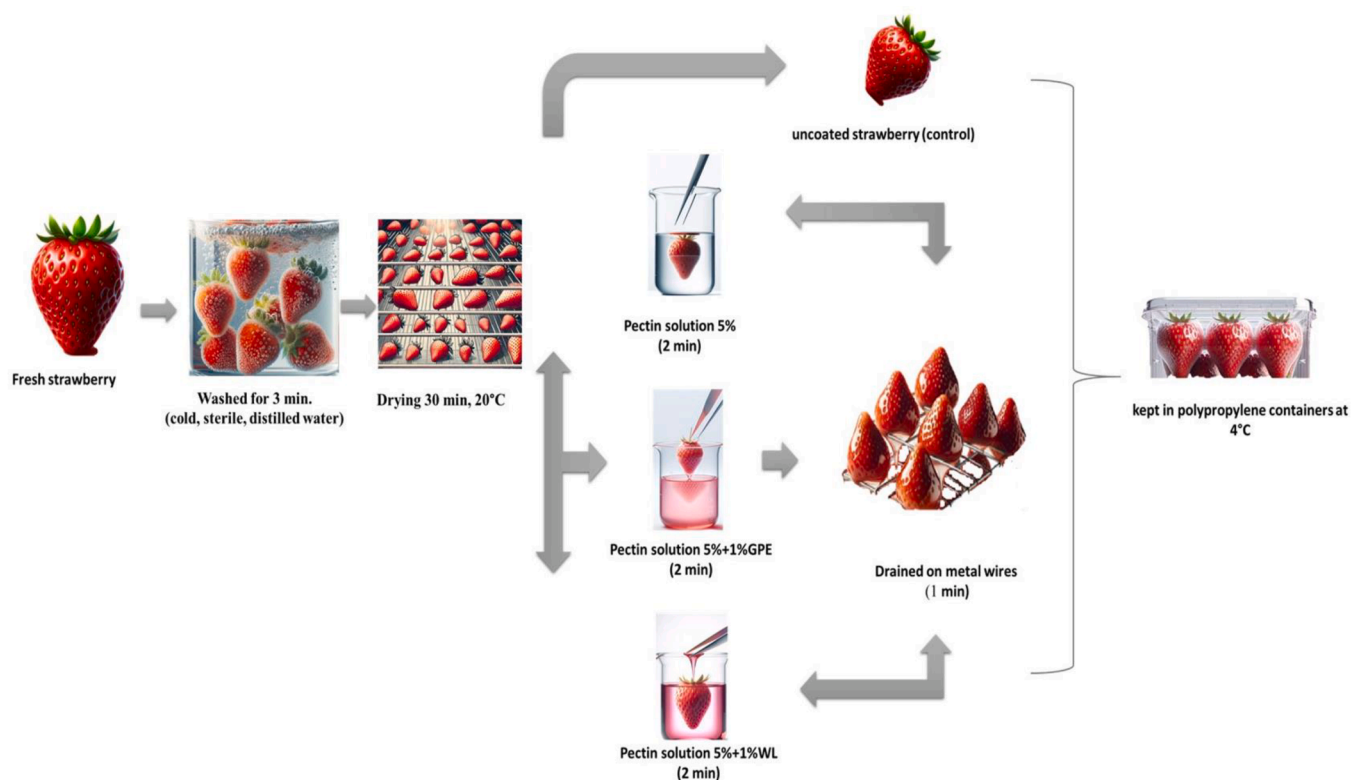


Fig. 1. Scheme of preparation of coating solutions and their application to strawberries.

HANNA Instruments, Woonsocket, RI, USA).

2.4.3. Surface color parameters

The color of the strawberries was evaluated using a CR-400 Chroma Meter (Konica Minolta Inc., Tokyo, Japan). L^* , a^* , and b^* chromaticity characteristics were used to define brightness, greenness/redness, and blueness/yellowness, respectively. The color change value (ΔE) was calculated using the following equation (Wang et al., 2020):

$$\Delta E = \sqrt{(L - L_0)^2 + (a - a_0)^2 + (b - b_0)^2}$$

L , a , and b are the color values of the strawberries at the end of the storage period, while L_0 , a_0 , and b_0 are the values on the initial day.

2.4.4. Firmness

The firmness of the strawberries was measured with a texture analyzer (TA-HD plus, Stable Micro Systems, Surrey, UK) with a P/36R probe and at 5 mm/min speed. The values were expressed as Newton (N). This analysis was carried out in 5 replicates (Kaynarca et al., 2023).

2.4.5. Microorganism count

Strawberries (10 g) were homogenized using a Stomacher (Interscience, Saint Nom la Br  t  che, France) in sterile peptone water. Total mesophilic aerobic bacteria (TMAB) and yeast-mold counts were determined on the 0, 5, 10, 15, and 20th day of storage. The TMAB count was measured using the surface plate technique on plate count agar (PCA, Merck, Darmstadt, Germany) after 48 h of incubation at 30 °C. Yeast mold counts were measured on potato dextrose agar (PDA, Merck) after 5 days of incubation at 25 °C. The microbiological counts were calculated as log colony forming units (cfu) per gram of material.

2.4.6. Antioxidant activity of coated strawberries

The antioxidant capacity of strawberries during storage was carried out as described under "Bioactive compounds and antioxidant capacity of coating solutions" in Section 2.3.

2.4.7. Bioactive compound of coated strawberries and degradation

The strawberries were extracted with 80 % methanol at a ratio of 1:5 over the storage period. The total phenolic and anthocyanin contents of the extracts were determined according to the protocol described under "Bioactive compound and antioxidant capacity of coating solutions" in Section 2.3. The TAC of the strawberries was calculated based on a molar absorptivity (ϵ) of 22400 L cm⁻¹ mg⁻¹ and molecular weight (MW) of 433.2 g mol⁻¹ of pelargonidin 3-glucoside.

The ascorbic acid content was determined by using the 2,6-dichlorophenol-indophenol (0.25 g/L) titrimetric test (Vahid, 2012). Each mL of 2,6-dichlorophenol-indophenol solution is equivalent to 0.2 mg ascorbic acid and the results were calculated as mg L⁻¹ ascorbic acid per 100 g strawberry.

The pseudo first-order kinetic rate constants (k) for each storage day was obtained by plotting a graph of $\ln(C_t/C_0)$ against time to estimate the kinetic degradation parameters of the bioactive components. The modeling equation is shown below where C_t is the concentration of the bioactive component at time t and C_0 is the initial concentration.

$$C = C_0 \exp(-kt)$$

2.5. Statistical analyses

Statistical analysis of the differences in the coating of strawberries with different solutions were carried out using Duncan multiple range tests and ANOVA using SPSS 17.0 software (SPSS Inc., Chicago, IL, US). All experiments were carried out in five replicates. 95 % was chosen as the statistical significance level.

3. Results and discussion

3.1. Weight loss

During storage, weight loss of the fruit can happen due to respiration, perspiration, and oxidation because of the thin skin and delicate tissue of

the fruit (Kaynarca et al., 2023; Pinzon et al., 2020). The weight loss percentage (% weight loss) of the control and coated strawberries is shown in Fig. 2-A. All fruit samples showed an increasing trend of weight loss during the storage period. The weight loss of pectin-coated samples showed a linear increase with storage time, while the uncoated sample showed a sharp increase after the 10th day. The coated sample (PW) containing WLE showed statistically significantly lower weight loss than the other samples ($p < 0.05$). The samples coated with PW and PG had a significantly lower % weight loss than the other samples as well as the control sample ($p < 0.05$). Weight losses of all strawberries were found to be between 2 and 4 %, 3–6 %, 4–8 %, and 6–10 % on the 5th, 10th, 15th and 20th days of storage, respectively. The pectin coating was effective in limiting the physiological deterioration of the fruit during storage by altering the surrounding atmosphere and changing the oxygen levels inside the fruit, thereby delaying the ripening process and reducing weight loss (Chandel et al., 2022). A remarkably low weight loss was observed in the current study over the 20-day storage period compared to previous reports in the literature (Benavides and Franco, 2023; Liu et al., 2021; Martínez-González et al., 2020; Pinzon et al., 2020; Riaz et al., 2021). Water loss, which causes changes in the appearance, color, and aroma of the fruit, accelerates aging, pathogen development and wrinkling in the product, leading to economic losses (De Bruno et al., 2023). The samples PW and PG showed less weight loss at the end of the storage period, most likely through an increase in the hydrophobic component of the biopolymer matrix through the phenolic components (WLE and GPE) and a decrease in

water vapor permeability (Noshirvani et al., 2023). The oxidative processes that cause spoiling were eventually slowed down by the pectin covering enhanced with bioactive ingredients, which restricts oxygen and moisture exchange.

3.2. pH, titratable acidity, and total soluble solids (TSS) content

The changes in titratable acidity, pH, and total soluble solids of strawberries during storage are shown in Table 1. The pH values of strawberries during the storage period ranged from 3.87 to 4.21. The highest pH was determined in the PW sample and the lowest pH was determined in the control sample at the start of the experiment ($p < 0.05$). On day 5, pH increased significantly in all samples except PW. On the 10th day, the pH of all samples with the bioactive coating showed an increase while the pH of the control samples without any bioactive coating remained constant, suggesting that the pectin coating contributed towards the increased the pH values of the samples. Similar outcomes were reportedly observed with strawberries covered in gum arabic containing bergamot essential oil (De Bruno et al., 2023). The fruits with extract-containing coatings appeared to show minimal pH change during storage compared to control and pectin-only coated samples, which may be related to the increased water loss observed. Changes in pH levels in fruits can be caused by the use of organic acids during respiration (Riaz et al., 2021).

The titratable acidity (TA) of the sample P at the start of the experiment was significantly low, whereas other samples yielded comparable

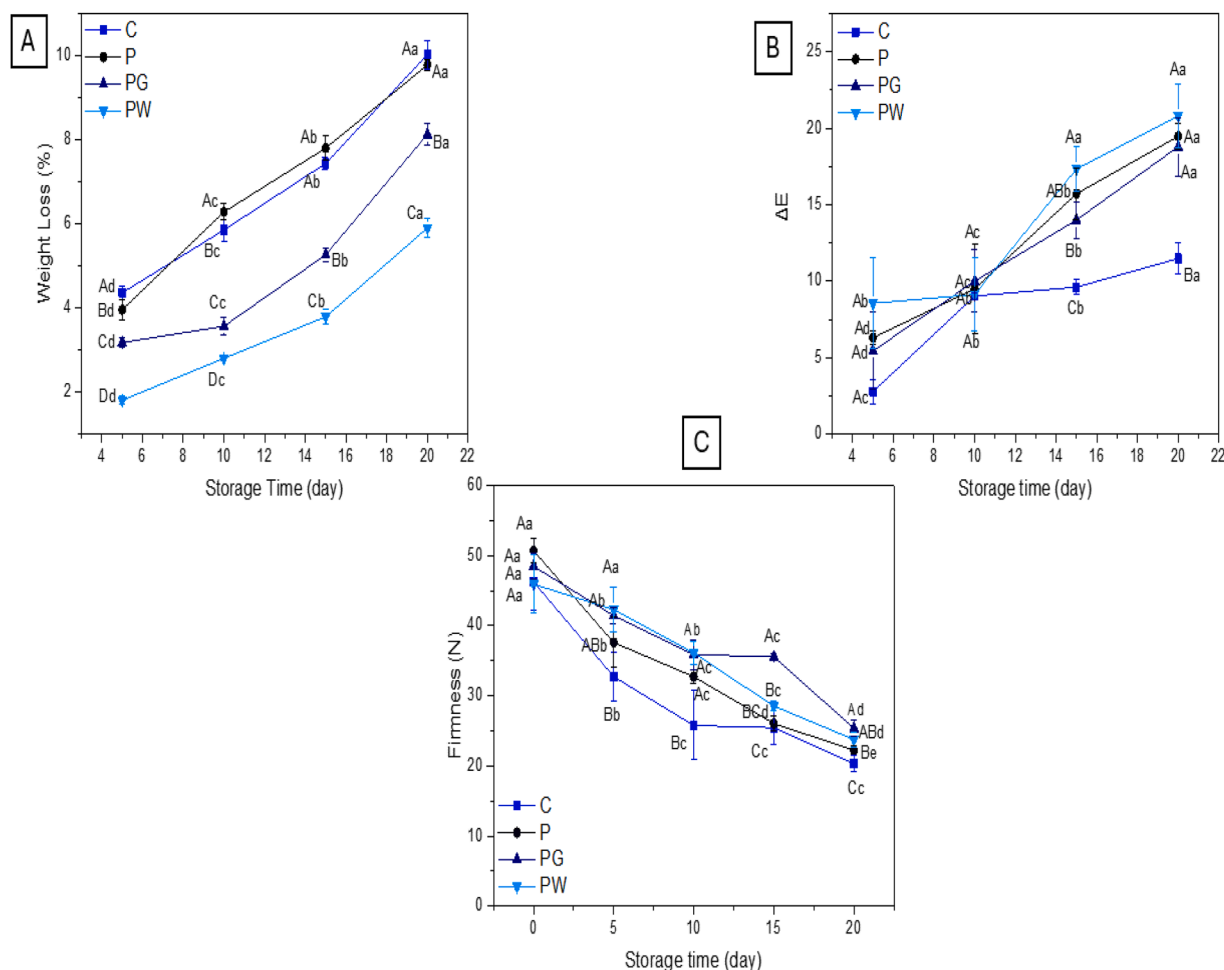


Fig. 2. Weight loss (A), color change (B), and hardness (C) of strawberries during storage time (in days). Values are given as means $\pm$ SD. Uppercase letters represent the difference between samples, lowercase letters represent the difference between storage times ($p < 0.05$). C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

Table 1

Titrateable acidity, pH, Total soluble solid content, and color parameters of uncoated and coated strawberries during storage time (in days).

		Storage Time (day)				
		0	5	10	15	20
Titrateable Acidity (g/100 g)	C	0.91 ±0.01 Aa	0.79 ±0.00 Ab	0.71 ±0.00 BCc	0.41 ±0.02 Cd	0.77 ±0.04 BCb
	P	0.83 ±0.06 Ba	0.71 ±0.02 Bb	0.69 ±0.02 Cb	0.56 ±0.00 Bc	0.73 ±0.02 Cb
	PG	0.86 ±0.00 ABa	0.79 ±0.02 Ab	0.73 ±0.02 ABc	0.58 ±0.02 Bd	0.87 ±0.00 Ba
	PW	0.87 ±0.02 ABa	0.77 ±0.05 Abc	0.75 ±0.00 Ac	0.75 ±0.02 Ac	0.81 ±0.02 Ab
	C	3.87 ±0.01 Ce	3.93 ±0.02 Cd	4.07 ±0.02 Cc	4.21 ±0.01 Aa	4.10 ±0.01 Ab
	P	3.96 ±0.01 Bc	4.00 ±0.01 Bb	4.10 ±0.01 Ba	4.01 ±0.01 Db	4.10 ±0.01 Aa
	PG	3.97 ±0.02 Bc	4.05 ±0.00 Ab	4.12 ±0.01 Aa	4.13 ±0.01 Ba	3.97 ±0.01 Cc
	PW	4.01 ±0.00 Ac	4.01 ±0.02 Bc	4.09 ±0.00 Ba	4.10 ±0.00 Ca	4.04 ±0.03 Bb
	C	11.05 ±0.05 Aa	9.65 ±0.35 Bb	9.15 ±0.15 Cc	9.30 ±0.10 Acd	8.85 ±0.15 Ad
	P	10.95 ±0.05 Aa	9.00 ±0.10 Ca	7.95 ±0.05 Db	7.85 ±0.05 Bbc	6.60 ±0.60 Bc
	PG	11.05 ±0.09 Aa	10.85 ±0.05 Aa	10.75 ±0.25 Aa	8.80 ±0.10 Cb	8.25 ±0.75 Ab
	PW	11.10 ±0.10 Aa	10.85 ±0.05 Aa	9.60 ±0.10 Bb	9.20 ±0.20 Abc	8.75 ±0.75 Ac
pH	C	38.93 ±1.43 Aa	37.00 ±1.56 ABa	33.31 ±1.52 Bb	32.59 ±0.12 Cb	32.08 ±1.79 Ab
	P	37.18 ±0.62 Aa	35.83 ±0.88 Bab	35.77 ±1.90 ABab	34.36 ±0.29 Bbc	33.59 ±1.01 Ac
	PG	38.86 ±0.44 Aa	34.94 ±1.53 Bb	34.63 ±0.32 Bb	33.09 ±0.34 Cc	33.00 ±0.35 Ac
	PW	38.55 ±0.66 Aa	38.49 ±0.29 Aa	38.03 ±1.24 Aa	37.97 ±0.66 Aa	34.33 ±1.43 Ab
	C	36.91 ±1.85 Ba	31.60 ±0.93 Ab	28.19 ±0.90 Bc	22.21 ±0.26 Bd	20.17 ±0.76 Be
	P	32.04 ±1.50 Ca	31.22 ±2.86 Aab	27.41 ±2.77 Bbc	27.28 ±1.93 Abc	26.72 ±1.39 Ac
	PG	38.42 ±0.33 Aa	35.18± 2.70 Aa	29.70± 1.89 Bb	27.60 ±1.84 Ab	22.10 ±1.83 Bc
	PW	41.24 ±2.17 Aa	34.29± 4.51 Ab	33.95± 2.57 Ab	27.10 ±1.92 Ac	25.08 ±1.87 Ac
	C	10.92 ±0.99 Bc	12.23± 1.25 Bc	15.07± 0.36 Ab	16.06 ±0.48 Bab	17.99 ±2.04 Ba
	P	13.84 ±0.69 Ac	16.48± 1.14 Cb	16.77± 1.28 Ab	18.44 ±1.03 Ab	22.84 ±1.53 Aa
	PG	10.45 ±0.98 Bc	10.03± 0.82 Bc	12.32± 1.42 Bb	17.05 ±1.11 ABa	17.59 ±0.26 Ba
	PW	7.49 ±0.69 Cd	11.89± 0.19 Ac	12.57± 1.06 Bc	17.34 ±1.70 ABb	19.79 ±0.97 Ba
Total Soluble Solid content (%)	C	11.05 ±0.05 Aa	9.65 ±0.35 Bb	9.15 ±0.15 Cc	9.30 ±0.10 Acd	8.85 ±0.15 Ad
	P	10.95 ±0.05 Aa	9.00 ±0.10 Ca	7.95 ±0.05 Db	7.85 ±0.05 Bbc	6.60 ±0.60 Bc
	PG	11.05 ±0.09 Aa	10.85 ±0.05 Aa	10.75 ±0.25 Aa	8.80 ±0.10 Cb	8.25 ±0.75 Ab
	PW	11.10 ±0.10 Aa	10.85 ±0.05 Aa	9.60 ±0.10 Bb	9.20 ±0.20 Abc	8.75 ±0.75 Ac
	C	38.93 ±1.43 Aa	37.00 ±1.56 ABa	33.31 ±1.52 Bb	32.59 ±0.12 Cb	32.08 ±1.79 Ab
	P	37.18 ±0.62 Aa	35.83 ±0.88 Bab	35.77 ±1.90 ABab	34.36 ±0.29 Bbc	33.59 ±1.01 Ac
	PG	38.86 ±0.44 Aa	34.94 ±1.53 Bb	34.63 ±0.32 Bb	33.09 ±0.34 Cc	33.00 ±0.35 Ac
	PW	38.55 ±0.66 Aa	38.49 ±0.29 Aa	38.03 ±1.24 Aa	37.97 ±0.66 Aa	34.33 ±1.43 Ab
	C	36.91 ±1.85 Ba	31.60 ±0.93 Ab	28.19 ±0.90 Bc	22.21 ±0.26 Bd	20.17 ±0.76 Be
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	PW	41.24 ±2.17 Aa	34.29± 4.51 Ab	33.95± 2.57 Ab	27.10 ±1.92 Ac	25.08 ±1.87 Ac
	C	10.92 ±0.99 Bc	12.23± 1.25 Bc	15.07± 0.36 Ab	16.06 ±0.48 Bab	17.99 ±2.04 Ba
L*	P	13.84 ±0.69 Ac	16.48± 1.14 Cb	16.77± 1.28 Ab	18.44 ±1.03 Ab	22.84 ±1.53 Aa
	PG	10.45 ±0.98 Bc	10.03± 0.82 Bc	12.32± 1.42 Bb	17.05 ±1.11 ABa	17.59 ±0.26 Ba
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	PG	10.45 ±0.98 Bc	10.03± 0.82 Bc	12.32± 1.42 Bb	17.05 ±1.11 ABa	17.59 ±0.26 Ba
	PW	7.49 ±0.69 Cd	11.89± 0.19 Ac	12.57± 1.06 Bc	17.34 ±1.70 ABb	19.79 ±0.97 Ba
	C	36.91 ±1.85 Ba	31.60 ±0.93 Ab	28.19 ±0.90 Bc	22.21 ±0.26 Bd	20.17 ±0.76 Be
	P	32.04 ±1.50 Ca	31.22 ±2.86 Aab	27.41 ±2.77 Bbc	27.28 ±1.93 Abc	26.72 ±1.39 Ac
	PG	38.42 ±0.33 Aa	35.18± 2.70 Aa	29.70± 1.89 Bb	27.60 ±1.84 Ab	22.10 ±1.83 Bc
	PW	41.24 ±2.17 Aa	34.29± 4.51 Ab	33.95± 2.57 Ab	27.10 ±1.92 Ac	25.08 ±1.87 Ac
	C	10.92 ±0.99 Bc	12.23± 1.25 Bc	15.07± 0.36 Ab	16.06 ±0.48 Bab	17.99 ±2.04 Ba
	P	13.84 ±0.69 Ac	16.48± 1.14 Cb	16.77± 1.28 Ab	18.44 ±1.03 Ab	22.84 ±1.53 Aa
	PG	10.45 ±0.98 Bc	10.03± 0.82 Bc	12.32± 1.42 Bb	17.05 ±1.11 ABa	17.59 ±0.26 Ba
	PW	7.49 ±0.69 Cd	11.89± 0.19 Ac	12.57± 1.06 Bc	17.34 ±1.70 ABb	19.79 ±0.97 Ba
b*	C	36.91 ±1.85 Ba	31.60 ±0.93 Ab	28.19 ±0.90 Bc	22.21 ±0.26 Bd	20.17 ±0.76 Be
	P	32.04 ±1.50 Ca	31.22 ±2.86 Aab	27.41 ±2.77 Bbc	27.28 ±1.93 Abc	26.72 ±1.39 Ac
	PG	38.42 ±0.33 Aa	35.18± 2.70 Aa	29.70± 1.89 Bb	27.60 ±1.84 Ab	22.10 ±1.83 Bc
	PW	41.24 ±2.17 Aa	34.29± 4.51 Ab	33.95± 2.57 Ab	27.10 ±1.92 Ac	25.08 ±1.87 Ac
	C	10.92 ±0.99 Bc	12.23± 1.25 Bc	15.07± 0.36 Ab	16.06 ±0.48 Bab	17.99 ±2.04 Ba
	P	13.84 ±0.69 Ac	16.48± 1.14 Cb	16.77± 1.28 Ab	18.44 ±1.03 Ab	22.84 ±1.53 Aa
	PG	10.45 ±0.98 Bc	10.03± 0.82 Bc	12.32± 1.42 Bb	17.05 ±1.11 ABa	17.59 ±0.26 Ba
	PW	7.49 ±0.69 Cd	11.89± 0.19 Ac	12.57± 1.06 Bc	17.34 ±1.70 ABb	19.79 ±0.97 Ba
	C	36.91 ±1.85 Ba	31.60 ±0.93 Ab	28.19 ±0.90 Bc	22.21 ±0.26 Bd	20.17 ±0.76 Be
	P	32.04 ±1.50 Ca	31.22 ±2.86 Aab	27.41 ±2.77 Bbc	27.28 ±1.93 Abc	26.72 ±1.39 Ac
	PG	38.42 ±0.33 Aa	35.18± 2.70 Aa	29.70± 1.89 Bb	27.60 ±1.84 Ab	22.10 ±1.83 Bc
	PW	41.24 ±2.17 Aa	34.29± 4.51 Ab	33.95± 2.57 Ab	27.10 ±1.92 Ac	25.08 ±1.87 Ac

Values are given as means ±SD. Upper letters represent the difference between samples, lowercase letters represent the difference between storage times ($p < 0.05$). C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

results ($p > 0.05$). TA values showed a decrease in all samples through the 15-day period. Acidity values are known to decrease due to the breakdown of organic acids with aging (Riaz et al., 2021). The TA of the pectin-WLE coated samples, however, remained stable between days 5 and 15, in contrast to the other samples suggesting that PW may be a suitable coating material. The TA showed an increase in all of the samples on the 20th day, which may be attributed to an increase in the fungal population in the samples and the resulting release of metabolites and acids (Riaz et al., 2021). The TA values were in accordance with the findings of De Bruno et al. (2023).

The TSS values of coated and uncoated strawberries were similar at the start of the experiment with values ranging from 11.10 to 10.95 ($p > 0.05$). Contrary to previous reports (Benavides and Franco, 2023; do Lago et al., 2023; Martínez-González et al., 2020; Pinzon et al., 2020), the TSS value in the current study was seen to be consistently higher. The control sample showed a reduction in TSS value as a result of storage; however, the TSS value of PG remained stable after 10 days. The other coated strawberries showed a decrease in TSS after the 5th day. A similar decrease during the storage period was reported by Gupta et al. (2023) and Riaz et al. (2021). The change in the TSS value was more modest in the coated samples than the control sample. Fruit ripening and respiration, as well as hydrolysis of carbohydrates and reduction of total sugars may all have contributed to a decrease in TSS (De Bruno et al., 2023).

3.3. Surface color parameters

Color is one of the most important parameters in consumer acceptance. Minimizing color losses by coating strawberries can also lead to less wastage and thereby mitigate economic losses (Pinzon et al., 2020). The color change (ΔE) value of the strawberries generally showed an increase during storage (Fig. 2-B). The difference in the ΔE of the products with coatings on different storage days was did not reach statistical significance ($p > 0.05$), while the ΔE value of the control sample was generally found to be lower. This may be due to the overall lower brightness (L^*) values of the coated samples. In the current study, the L^* value showed a trend for decrease with increase in the storage period in all samples, corroborating the data of De Bruno et al. (2023). At the end of 15 days, the L^* value of PW showed a significant decrease, suggesting that this sample was the most stable. Strawberries coated with *Aloe vera* gel (Mohammadi et al., 2021) and chitosan (Martínez-González et al., 2020) also exhibited similar L^* values. Coatings containing grape waste extract showed a significant increase in the redness (a^*) value (Table 1). The a^* value of the control sample decreased gradually with storage; the coated samples showed a comparatively less decrease. The red color of strawberries is attributed to the presence of anthocyanins. The decrease in the redness of strawberries is due to the deterioration of anthocyanin during storage (Gupta et al., 2023). In terms of the a^* value, PG exhibited the highest degree of stability, followed by PW, suggesting that the extracts can increase the stability of the color and thereby the appeal of strawberries. Contrary to the a^* value, the yellowness (b^*) value increased in all samples with storage time. The PG sample had the lowest b^* value and the lowest rate of growth.

3.4. Firmness

The differences in firmness values of the coated and control strawberry samples over the period of 20 days of cool storage are shown in Fig. 2-C. No significant difference was identified in the firmness values of coated and uncoated samples on the first day of storage ($p > 0.05$). The uncoated control sample showed the lowest firmness value on the

5th day ($p < 0.05$), decreasing from 46 to 50 N at the start of the experiment to 20–25 N at the end of 20 days. A high water content of the fruit can contribute to its hardness due to turgor pressure; thus, the hardness tends to decrease with water loss (Benavides and Franco, 2023). Uncoated strawberries were shown to exhibit increased water loss and fungal contamination, as well as significant tissue senescence and ruptured cell walls (Riaz et al., 2021). The PG sample had the smallest decrease in hardness, followed by PW. The presence of GPE and WLE in the coating could therefore effectively conserve firmness by inhibiting water loss, surpassing the performance of other sample groups. The addition of GPE and WLE to the coatings might enhanced the porosity and hydrophobicity of the pectin films, resulting in a harmonious combination of moisture retention and gas permeability (Thakur et al., 2018). This balance may have stabilised the moisture barrier without hindering gas exchange, which is essential to maintain fruit quality, minimise dehydration and preserve volatile aromatics. These data are also supported by the weight loss results observed in the current study. Coatings containing pectin (Adhikari et al., 2023; Treviño-Garza et al., 2015) and plant extracts (Benavides and Franco, 2023; De Bruno et al., 2023; Mohammadi et al., 2021; Riaz et al., 2021) were all reported to show similar outcomes.

3.5. Microorganism counts

Strawberries are highly susceptible to tissue injury and infection caused by various phytopathogenic fungi, such as *Botrytis cinerea*, *Rhizopus stolonifer*, and *Penicillium* spp. (Pinzon et al., 2020). The microbial count of uncoated and coated strawberries during cold storage are illustrated in Fig. 3 and strawberry pictures taken in real time during storage are given in Fig. 4. The initial total mesophilic aerobic bacteria (TMAB) count of strawberries was 2.50–3.20 log cfu/g. The control sample had the highest TMAB value, whereas strawberries with the coatings showed the lowest values on the initial day. The TMAB count of the control sample had a 46 % increase on day 5, whereas the coated samples exhibited an approximate increase of 35 %. The TMAB count showed a positive correlation with the duration of storage, as also corroborated by Hassanein et al. (2024); Kaynarca et al. (2023); Nosh-irvani et al. (2023) and Yıldırım-Yalçın et al. (2022).

At the end of the 20th day, the lowest TMBA value was determined in the PG sample with 5.19 ± 0.09 log cfu/g, suggesting that the grape pomace extract was more successful in keeping low TMBA counts. Polyphenols present in grape pomace were found to have a bactericidal effect against gram-negative bacteria (Kabir et al., 2015; Mollica et al.,

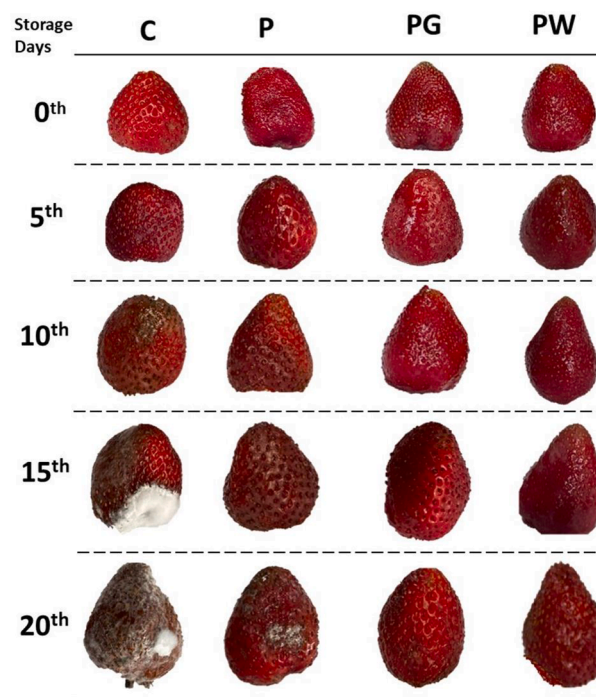


Fig. 4. Strawberry images taken in real time during storage. C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

2021). Wine lees also have higher antibacterial activity against Gram-negative bacteria such as *Escherichia coli* and *Pseudomonas aeruginosa* than Gram-positive bacteria (Tagkouli et al., 2022).

The total mold and yeast (MY) count in the C, P, PG, and PW samples on Day 1 of storage were detected as 4.07 ± 0.12 , 3.72 ± 0.17 , 3.36 ± 0.10 , and 3.11 ± 0.11 log cfu/g, respectively. The samples exhibited a reduction in the MY count on the 5th day of storage, that coincided with an increase in the TMAB count. On day 5, the MY count in the control sample exhibited a considerably greater value compared to the coated samples, suggesting that the pectin coating may reduce the MY count. The capacity of the pectin coating to inhibit microbial growth can be attributed to the presence of a galacturonic acid biopolymer that lowers the pH and enhances barrier properties that can prevent oxygen from

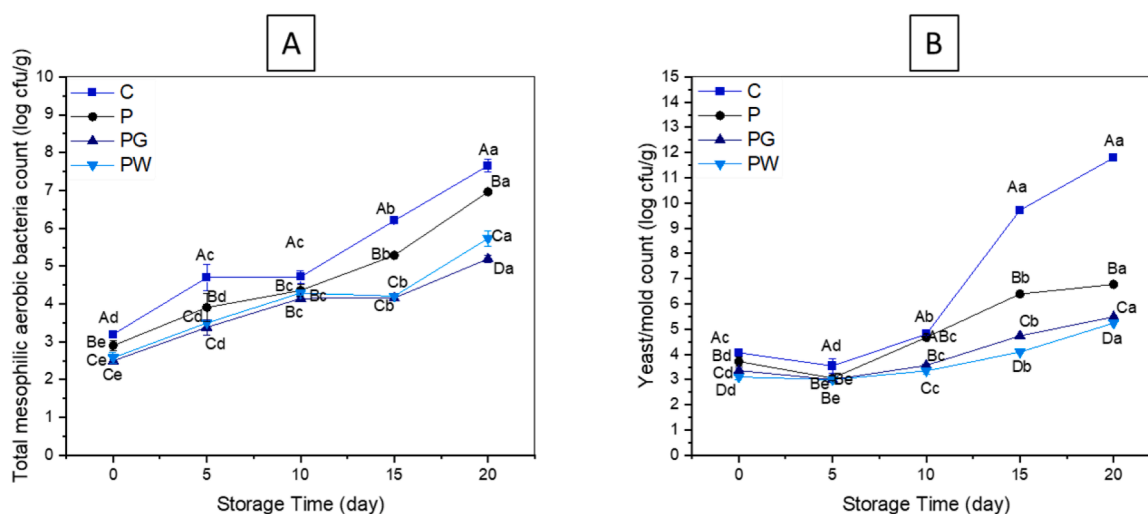


Fig. 3. Microbial count of control and coated strawberries during storage time. Values are given as means $\pm$ SD. Uppercase letters represent the difference between samples, lowercase letters represent the difference between storage times ($p < 0.05$). C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

reaching the microorganisms (Dhawan and Chakraborty, 2024). The MY count was reported to show a slow increase in *Aloe vera* gel-coated strawberries (Sogvar et al., 2016) and pectin-coated fresh apple samples (Naqash et al., 2022) during the storage period, similar to the findings of the current study. A statistically significantly lower MY count in the coated samples containing bioactive components was observed beginning on the 10th day of storage, with the WLE containing coating showing significantly more suppression. Extracts containing phenolics from Cabernet Sauvignon, Carmenere, and Syrah grapes have been shown to effectively suppress the growth of the phytopathogenic fungus, *B. cinerea* (Mendoza et al., 2013). At the end of storage, MY number of control and pectin-coated samples (P, PG, PW) was determined as 11.79 ± 0.02 , 6.78 ± 0.02 , 5.49 ± 0.02 , and 5.24 ± 0.06 log cfu/g, respectively.

3.6. Antioxidant activity of coated strawberries

The antioxidant activities of strawberries showed a decrease with increased storage time (Table 2), which may be attributed to the formation of free radicals during the storage period (HosseiniFarahi et al., 2020). The antioxidant capacities of bioactive coated strawberries were significantly greater, particularly in the presence of WLE and GPE. DPPH• scavenging activities (IC₅₀) of the coating solutions were 1.08 ± 0.05 (Pectin), 0.86 ± 0.03 (PG), and 0.95 ± 0.02 (PW). The highest TEAC value was found in the PG coating solution (5.66 ± 0.62 μmol Trolox/kg), followed by PW (4.34 ± 0.13 μmol Trolox/kg) and P (2.52 ± 0.29 μmol Trolox/kg). Wathoni et al. (2019) reported that the antioxidant activity in pectin obtained from Indonesian mangosteen (*Garcinia mangostana* L.) rind could be attributed to the hydroxyl groups of polysaccharides when the viscosity was not too high. At the end of 20 days of storage in the current study, the amount of coating material required to inhibit the DPPH radical by 50 % (IC₅₀) increased by 40 % in the C, P and PG samples, and by 26 % in the PW sample. Although the IC₅₀ value in the PW sample remained unchanged after the 10th day of storage, it continued to progressively rise in the other samples. The TEAC values of all samples, except PW, exhibited a decline with time, similar to the DPPH• scavenging activity. The TEAC value of the PW sample showed a decrease of only 13 % at the end of storage. Benavides and Franco (2023) suggested that the antioxidant activity of arrayan (*Luma apiculata*) extract chitosan-coated strawberries decreased by approximately 16 % at the end of 21 days, and 70 % in control samples. The PW coating may have reduced oxidation and aging by protecting the antioxidant

compounds by decreasing gas exchange (HosseiniFarahi et al., 2020). Thus, a combinatorial effect was observed with the use of Pectin and WLE on the preservation of antioxidant activity.

3.7. Degradation of the bioactive compounds in coated strawberries

The current study analyzed the variations in bioactive components, including total phenolic compounds, total anthocyanins, and ascorbic acid in strawberries that were both uncoated and coated with pectin, PG, or PW (Table 2). The degradation of these components during the preservation period is shown in Fig. 5. The total phenolic content (TPC) was analyzed to examine the contribution of coating solutions. The control sample had the lowest TPC (342.07 mg GAE/kg), while PG had the highest (410.70 mg GAE/kg), followed by PW (397.97 mg GAE/kg). As the storage time increased, the TPC value in the samples gradually decreased. Nonetheless, the TPC values of strawberries coated with pectin containing bioactive components were significantly higher than the other samples. TPC was not detected in pectin; but was determined as 6.64 ± 1.66 mg GAE/kg and 8.41 ± 1.11 mg GAE/kg in PG and PW coatings, respectively. The control sample had a TPC retention rate of 55 %, while the same was 60 % in the PG sample and 66 % in the PW sample after a storage period of 20 days. The decrease in TPC seen during fruit storage might be ascribed to the disintegration of cell structure as the fruit ages (Kaynarca et al., 2023). The TPC may also have decreased as a result of oxidation processes (Ali et al., 2019). Martínez-González et al. (2020) reported a similar reduction in chitosan/propolis-coated strawberries; however, an increase in the propolis extract concentration did not decrease the loss of TPC. We also examined the losses in TPC during the storage period of strawberries in a kinetic analysis. A plot of the logarithm of TPC retention against time (Fig. 5-A) indicated the presence of stable straight lines ($R^2=0.96-0.88$). Total phenolic degradation of coated samples containing WLE and GPE was less during the storage period. Accordingly, the TPC degradation reaction rate constant (k) was determined as $3.18 \cdot 10^{-2}$, $3.33 \cdot 10^{-2}$, $1.99 \cdot 10^{-2}$, and $1.64 \cdot 10^{-2}$ day⁻¹ for C, P, PG and PW, respectively. The fact that the PW degradation rate constant was 50 % lower than the control group suggested that the addition of WLE to the coating may have the potential to delay the degradation of phenolic substances without contributing to the storage time of the strawberries. *Aloe vera* gel coatings containing 2 mM salicylic acid also showed similar degradation pattern as our study compared to the control strawberry samples (HosseiniFarahi et al., 2020).

Table 2
Bioactive components and antioxidant activity of uncoated and coated strawberries during storage.

		Storage Time (day)				
		0	5	10	15	20
DPPH• scavenging activity, IC50* (mg/L)	C	0.078±0.001 Ac	0.089±0.002 Ab	0.089±0.002 Ab	0.104±0.002 Aa	0.110±0.008 Aa
	P	0.078±0.002 Ad	0.080±0.002 Bcd	0.082±0.002 Bc	0.091±0.001 Cb	0.103±0.001 Aa
	PG	0.070±0.001 Be	0.076±0.000 Cd	0.091±0.002 Ac	0.097±0.002 Bb	0.112±0.005 Aa
	PW	0.071±0.001 Bc	0.087±0.001 Ab	0.090±0.001 Aa	0.090±0.001 Ca	0.090±0.001 Ba
	C	41.77± 0.25 Ca	35.29± 2.90 Bb	33.88± 1.89 Bbc	30.70± 0.54 BCcd	29.87± 2.28 BCd
TEAC (μmol Trolox/kg)	P	43.34±0.62 Ba	36.94±0.99 Bb	35.58±0.045 Bc	32.19±0.04 Bd	30.12±1.12 Be
	PG	48.88±0.62 Aa	36.28±0.17 Bb	31.20±0.54 Cc	29.25±1.24 Cd	27.35±0.75 Ce
	PW	48.51±0.66 Aa	47.56±2.03 Aa	43.07±0.70 Ab	43.37±1.70 Ab	42.64±0.91 Ab
	C	342.07±3.88 Da	312.73±0.00 Cb	276.75±1.66 Cc	254.61±1.66 Bd	189.15±2.77 De
Total Phenolic Content (mg GAE/kg)	P	384.69±6.65 Ca	326.57±0.56 Bb	295.02±9.96 Bc	257.38±6.64 Bd	196.35±0.00 Ce
	PG	410.70±1.66 Aa	335.98±8.86 Ab	303.32±6.09 Bc	290.04±2.77 Ad	244.65±4.98 Be
	PW	397.97±6.64 Ba	334.32±1.66 ABb	317.16±4.43 Ac	292.25±1.66 Ad	261.25±1.67 Ae
	C	222.01±2.33 Ca	172.89±1.16 Cb	121.06±1.94 Dc	87.03±0.39 Dd	49.12±8.90 De
Total anthocyanin content (mg /kg)	P	224.35±2.33 Ca	177.53±2.54 Cb	140.02±0.78 Cc	98.24±3.87 Cd	73.88±2.71 Ce
	PG	234.78±1.94 Ba	186.82±2.71 Bb	167.86±1.55 Bc	120.68±1.55 Bd	100.95±1.16 Be
	PW	252.96±1.55 Aa	218.15±3.87 Ab	177.53±6.58 Ac	155.10±1.16 Ad	133.33±3.12 Ae
	C	369.23±0.00 Aa	276.92±0.00 Cb	213.13±0.00 Ac	150.77±32.64 Cd	131.36±0.00 Ad
Ascorbic Acid Content (mg/100 g)	P	334.61±11.54 Aa	305.12±14.58 Bbc	276.92±0.00 Ac	164.55±54.85 BCd	144.14±0.00 Ad
	PG	375.00±0.00 Aa	334.46±4.92 Ab	302.22±0.00 Ac	257.69±19.23 Ad	193.08±0.00 Ae
	PW	345.45±39.40 Aa	328.64±4.09 Ab	234.10±0.00 Ac	219.40±0.00 ABcd	190.4 ± 0.00 Ad

Values are given as means ±SD. Uppercase letters represent the difference between samples, lowercase letters represent the difference between storage times (*p* < 0.05). C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

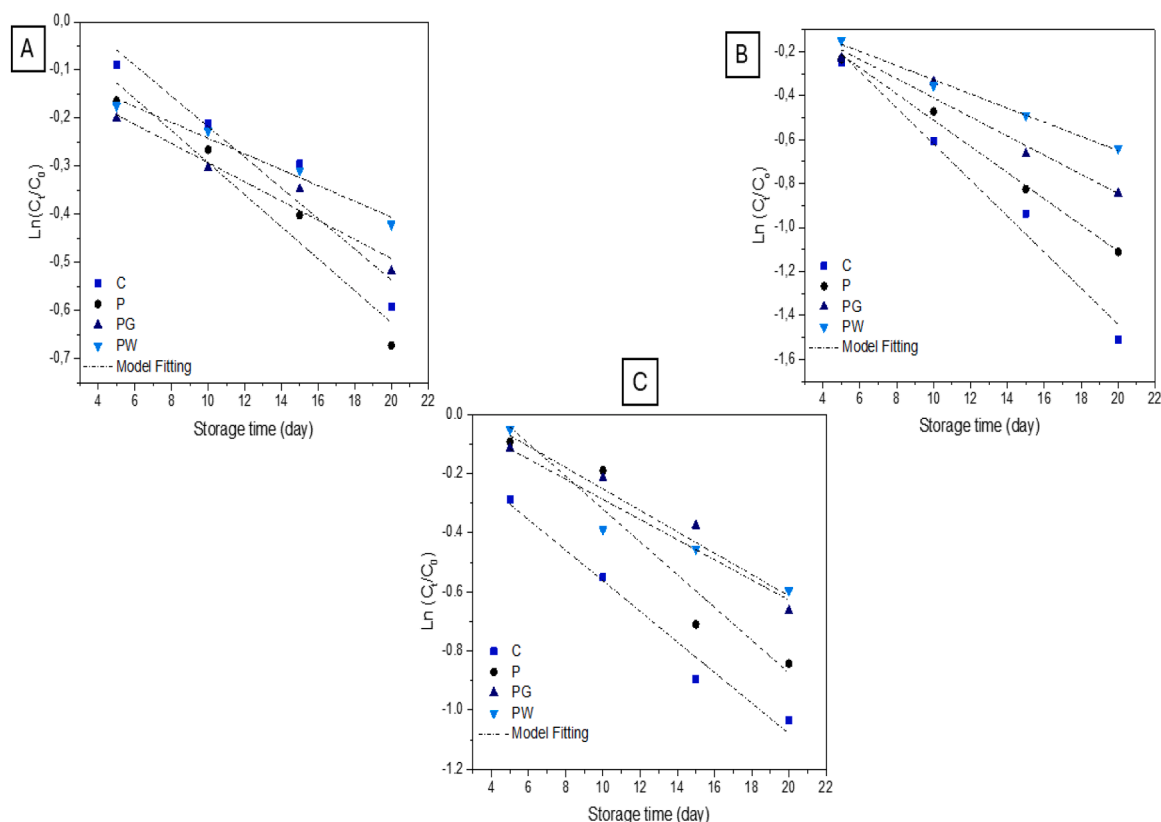


Fig. 5. Degradation total phenolic content (A), total anthocyanin content (B), and ascorbic acid content (C) of control and coated strawberries during storage. C: uncoated strawberries, P: pectin coated strawberries, PG: pectin-GPE coated strawberries, PW: pectin-WLE coated strawberries.

The red color of strawberries, which determines their ripeness and quality, is due to the presence of water-soluble anthocyanins (Mustafa et al., 2022). Anthocyanins, like phenolic compounds and ascorbic acid, are valuable compounds because they protect cells against oxidative stress (HosseiniFarahi et al., 2020). Control pectin coating solutions lacked the presence of anthocyanins. The total anthocyanin content (TAC) of PG and PW solutions was determined as 1.06 ± 0.35 and 4.94 ± 0.00 mg malvidin-3-glucoside/kg, respectively. The sample with the highest TAC value (252.96 ± 1.55 mg pelargonidin 3-glucoside/kg) was PW. The lowest values were observed in the samples C and P. The TAC value showed a continuous decrease with storage time in all samples, which could be attributed to exposure to oxygen over time (Benavides and Franco, 2023). The decrease in anthocyanin in coated fruits may also be attributed to an increase in the activity of enzymes such as polyphenol oxidase (PPO) (Riaz et al., 2021). The decline in anthocyanins the PW and PG samples was relatively minor compared to the other two samples (C and P). TAC retention in control strawberries was 22 % at the end of the storage period, compared to 43 % and 53 % in PG and PW samples, respectively. The lowest TAC value was found in the control sample and while the highest was in the PW sample during the storage period ($p < 0.05$). Although the overall decline was modest, the most pronounced decline in the TAC in the coated samples was in the 10th to 15th days of storage. The compliance of anthocyanin degradation with the first-order kinetic equation was proven by obtaining high R^2 values ranging from 0.94 to 0.99 (Fig. 5-B). TAC degradation showed an increasing trend with storage time in all samples. The highest TAC degradation reaction rate constant (k) was $8.22 \cdot 10^{-2} \text{ day}^{-1}$ in the control sample, while it was $3.22 \cdot 10^{-2} \text{ day}^{-1}$ in the PW sample. The k value of the control sample was 2.6 times higher than the PW sample and 1.9 times higher than the PG sample. The degradation of TAC was decreased by 46 % with PW coating and 27 % with PG coating. The findings indicate that WLE can be a more effective component than GPE,

making it a suitable coating material for strawberries.

The ascorbic acid (AA) content of all strawberry samples was similar at the start of the storage period ($p > 0.05$). All of the samples exhibited a decrease in AA with storage, similar to other bioactive substances. In general, this decrease slowed down between the 15th and 20th days of storage. The AA content of the control sample declined from 75 % on the 5th day of storage to 35 % on the 20th day. The AA content decreased from 85 % to 51 % in the PG sample, and from 95 % to 55 % in the PW sample over the same duration. Similar decreases were detected in plums coated with pectin-based and chitosan-based edible coatings (Benavides and Franco, 2023; Panahirad et al., 2020). Such a decline in AA levels during postharvest storage may be attributed to activities of the enzymes ascorbic-acid oxidase and PPO that are directly influenced by the oxygen levels in the surrounding environment (Panahirad et al., 2020). The rate of AA degradation could be fit to first-order kinetic equation as demonstrated by high R^2 values ranging from 0.90 to 0.98 (Fig. 5-C). The degradation reaction rate constant (k) was calculated as $5.16 \cdot 10^{-2}$, $5.54 \cdot 10^{-2}$, $3.62 \cdot 10^{-2}$, and $3.40 \cdot 10^{-2} \text{ day}^{-1}$ in the C, P, PG, and PW. Coatings with bioactive substances provided superior protection for AA. Accordingly, the lower degradation of phenolic compounds in PW and PG coated samples may be due to reducing effect of ascorbic acid, which is reversibly converted to dehydroascorbic acid (Lo Lago et al., 2023).

4. Conclusion

The current study investigated the possibility of utilizing pectin, a readily available and cost-effective resource, to extend the shelf life of strawberries. Furthermore, the current study suggested that the application of pectin-based coatings containing extracts of grape pomace (GPE) and wine lees (WLE) could potentially extend the shelf life of strawberries. Pectin films, enriched with GPE and WLE, known for their

high levels of bioactive components, effectively reduced the degradation of phenolic compounds, anthocyanins, and ascorbic acid in strawberries during storage. Samples coated with PW showed a minimal (12 %) decrease in antioxidant activity after 20 days in storage. PW coating offers the potential to significantly reduce free radical production, which could have beneficial effects on human health. Coating strawberries with PG and PW inhibited the growth of yeast and mold. Among all 4 samples tested, the PW coated samples had the best weight loss outcome. Overall, the current study shows that pectin-based coatings, enriched with grape pomace and wine lees extract rich in bioactive components, can play an important role in extending the shelf life of strawberries. Such repurposing of waste products for shelf-life extension can contribute to a decrease of food loss during both the transfer of products and their storage.

CRediT authorship contribution statement

Gulce Bedis Kaynarca: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Deniz Damla Altan Kamer:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Emel Yucel:** Methodology, Formal analysis, Data curation, Conceptualization. **Oylum Simal Yilmaz:** Methodology, Formal analysis, Conceptualization. **Yasemin Henden:** Formal analysis, Validation, Writing – original draft. **Elif Kaymaz:** Validation, Methodology, Formal analysis. **Tuncay Gumus:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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